

Resources

REAL VARIABLES I

Fall 2026

For every topic we cover, I will provide either lecture notes or a precise reading reference. The books below offer complementary treatments; you do not need to purchase them.

0.1 Main textbook

- Luigi Ambrosio, Giuseppe Da Prato, and Andrea Menzucchi, *Introduction to Measure Theory and Integration*, Publications of the Scuola Normale Superiore 10, Edizioni della Normale, 2011. This is our principal reference. Publisher page and electronic access.

0.2 Additional references

- Elias M. Stein and Rami Shakarchi, *Real Analysis: Measure Theory, Integration, and Hilbert Spaces*, Princeton Lectures in Analysis III, Princeton University Press, 2005. Book record.
- Sheldon Axler, *Measure, Integration & Real Analysis*, Graduate Texts in Mathematics 282, Springer, 2020. The author provides a legal, free electronic edition. Open-access book.
- Gerald B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Wiley, 1999. This is a more advanced reference with extensive coverage beyond the course. Publisher page.

0.3 Office hours

Office hours are part of the course. Bring questions about definitions, proofs, examples, reading, or homework - even when your question is only "where should I start?"

Discussing homework in office hours is encouraged. If you give a clear, complete solution of a designated homework exercise, I may propose to you a grade for that exercise during the meeting. If you are satisfied with the grade, you will not need to submit a written solution for that exercise.

0.4 Access and well-being

- Center for Educational Access
- Counseling and Psychological Services
- U of A Cares
- University emergency information
- RazALERT